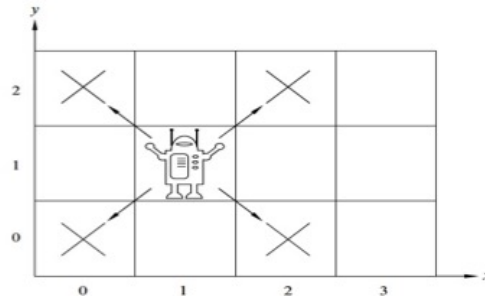


## Structural induction example robot grid



**Theorem:** A robot on an infinite 2-dimensional integer grid starts at  $(0,0)$  and at each step moves to diagonally adjacent grid point. This robot can / cannot (*circle one*) reach  $(1,0)$ .

**Definition** The set of positions the robot can visit  $Pos$  is defined by:

Basis Step:  $(0,0) \in Pos$

Recursive Step: If  $(x,y) \in Pos$ , then

are also in  $Pos$

*Example elements of  $Pos$  are:*

**Lemma:**  $\forall (x,y) \in Pos$  ( $x+y$  is an even integer)

*Why are we calling this a lemma?*

Proof of theorem using lemma: To show is  $(1,0) \notin Pos$ . Rewriting the lemma to explicitly restrict the domain of the universal, we have  $\forall (x,y) ((x,y) \in Pos \rightarrow (x+y \text{ is an even integer}))$ . Since the universal is true,  $((1,0) \in Pos \rightarrow (1+0 \text{ is an even integer}))$  is a true statement. Evaluating the conclusion of this conditional statement: By definition of long division, since  $1 = 0 \cdot 2 + 1$  (where  $0 \in \mathbb{Z}$  and  $1 \in \mathbb{Z}$  and  $0 \leq 1 < 2$  mean that 0 is the quotient and 1 is the remainder),  $1 \bmod 2 = 1$  which is not 0 so the conclusion is false. A true conditional with a false conclusion must have a false hypothesis:  $(1,0) \notin Pos$ , QED.  $\square$

Proof of lemma by structural induction:

**Basis Step:**

**Recursive Step:** Consider arbitrary  $(x, y) \in Pos$ . To show is:

$(x + y \text{ is an even integer}) \rightarrow (\text{sum of coordinates of next position is even integer})$

Assume as **the induction hypothesis, IH** that: